

- 3 (a)(i) An ideal gas is one that obeys the ideal gas equation, $pV = nRT$ at all pressures, volumes and temperatures. B1

(ii) $pV = nRT$

$$(1.00 \times 10^5) (750 \times 10^{-6}) = n (8.31) (300)$$

$$n = 0.030$$

M1

A1

(b)

	work done on gas / J	heat supplied to gas / J	increase in internal energy of gas / J
Ao B	+360	0	+360
B to C	0	+670	+670
C to D	-810	0	-810
D to A	0	-220	-220

Minus 1 per mistake.

- (c) the gas molecules bounce off the receding piston at lower speeds

And hence lower kinetic energy.

B1

For an ideal gas, the temperature is proportional to the average kinetic energy of the molecules.

B1

- (d) The net work by the engine is positive and can be used to move the car (turn the wheels)

B1

